6 **(a)** An image of object O is formed by a plane mirror.

A person views the image. Figure 6.1 shows a scale diagram of the arrangement.

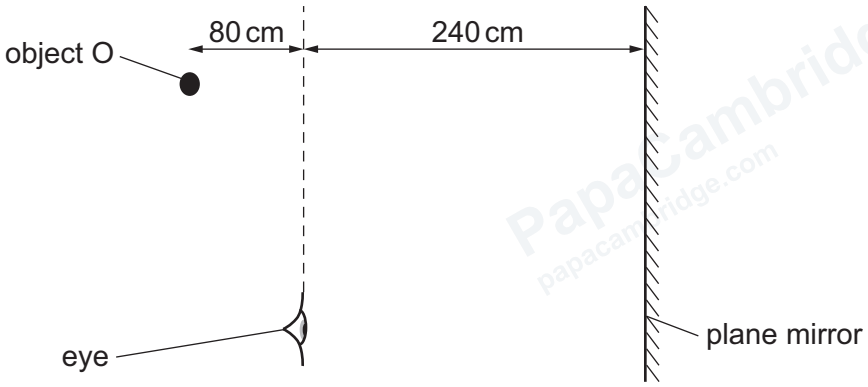

Figure 6.1

(i) Mark the position of the image. Label it I. [1]

(ii) Draw on Figure 6.1 to show the path of the light from object O to the person's eye. [2]

(b) Figure 6.2 shows three rays of light travelling from a light source S. The rays travel in glass towards a glass–water boundary.

NOT TO SCALE

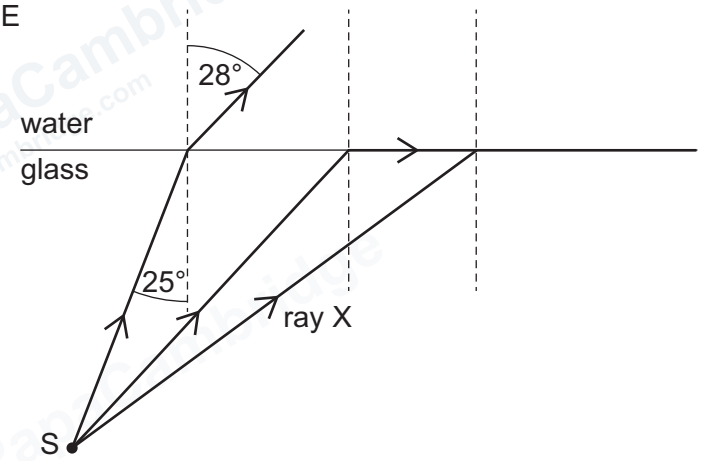

Figure 6.2

(i) Calculate the refractive index for the glass–water boundary. Use information from Figure 6.2.

refractive index = ... [2]

(ii) On Figure 6.2, label the critical angle *c*. [1]

(iii) On Figure 6.2, continue the path of ray X after it reaches the glass–water boundary. [1]

[Total: 7]

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